

Exp No: 13

Embedded c program to add an array of 16-bit numbers

And store the 32-bit result in internal ram

AIM:

To write an Embedded C program to add an array of 16-bit numbers and store the results in internal RAM using Keil software.

SOFTWARE REQUIRED:

- Keil

PROGRAM:

```
#include <reg51.h>
```

```
void main(void)
```

```
{
```

```
    unsigned int i;
```

```
    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
```

```
    unsigned long sum = 0;
```

```
    for (i = 0; i < 5; i++)
```

```
    {
```

```
        sum = sum + array[i];
```

```
    }
```

```
    P0 = (unsigned char)(sum & 0xFF); // Lower 8 bits
```

```
    P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
```

```
    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
```

```
    while(1); // keep output stable
```

```
}
```

OUTPUT:

The screenshot displays the Keil uVision IDE interface for a C program. The main window shows the source code of a program that calculates the sum of an array and outputs the result in three 8-bit segments. The registers window on the left shows the current state of the processor registers. The command window at the bottom left shows the load command. The call stack window at the bottom right shows the current function is MAIN.

Registers:

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
SP	0x15
PC	0x0000
PC	0x0909
SR	0x00194
PSW	0x00

Source Code:

```
3: void main(void)
4: {
5:     unsigned int i;
6:     unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
7:     unsigned long int sum = 0;
8:     for (i = 0; i < 5; i++)
9:     {
10:        sum = sum + array[i];
11:    }
12:    P0 = (unsigned char)(sum & 0xFF); // Lower 8 bits
13:    P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
14:    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
15:    while (1)
16:    {
17:        // Keep the output stable
18:    }
19: }
```

Command Window:

```
Running with Code Size Limit: 2K
Load "C:\Users\SUPRIYA V\OneDrive\ECA1415\Objects\exp-13"
```

Call Stack - Locals:

Name	Location/Value	Type
MAIN	0x0909	

Simulation Status:

Simulation: t1: 180.73402800 sec L6 C:1 CAP: NUM SCRL: OVR: R/W

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